ic_autolab round 37 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-10T04:09:00Z at git `7242133d0d8d7d54321bd2954f12a85398aa4475`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: none yet. Record one with `python3 tools/pcc/pcc.py claim new` before citing this round; until then its outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20 **null**	tw2l-x20-j16-n2	0.986 [0.941, 1.034]	0.965 [0.920, 1.011]	0.866 [0.824, 0.910]	sel < inc	2.469 / 2.649 / 4.263	2.503 / 2.746 / 4.925	1.538 / 1.649 / 2.645	1.6x / 1.6x / 1.6x
28 **null**	tw2l-x20-j16-n2	0.967 [0.927, 1.008]	0.968 [0.928, 1.010]	0.976 [0.935, 1.018]	sel < inc	1.834 / 2.011 / 3.605	1.897 / 2.077 / 3.695	1.005 / 1.099 / 1.945	1.8x / 1.8x / 1.9x
32 **improvement**	tw2l-x20-j16-n2	0.915 [0.864, 0.969]	0.897 [0.846, 0.951]	0.825 [0.775, 0.879]	sel < inc	1.715 / 1.884 / 3.411	1.875 / 2.101 / 4.132	0.966 / 1.061 / 1.915	1.8x / 1.8x / 1.8x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 20: compare ops.status = ok, admissible = true; sel / inc = 0.9863 [0.941, 1.034] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.9667 [0.927, 1.008] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 32: compare ops.status = ok, admissible = true; sel / inc = 0.9146 [0.864, 0.969] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 20: session `ECBS1hac1e089035db` (run exit 0, verify exit 0, receipt sha256 `f0f307de2133274b5e84be3d62587feb69c04d6c276f51eff7d2e2084bd8bcb2`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	16	16	1.703 / 1.830 / 2.972	matched reference
`inc`	wsum2g-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	16	16	2.454 / 2.691 / 4.822	incumbent
`tw2l-x20-j16-n2`	tw2l-x20-j16	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=2.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run`	16	16	2.307 / 2.470 / 3.940	the SNC-panel arm (PCC-20261110-978c29: 0.936 / 0.909 / 0.922 of the incumbent at k = 20 / 28 / 32 over 480 multi-curve pairs); round 36's 64-pair holdout left these sizes null (0.998 / 0.958 / 0.942 with intervals touching 1), so this round re-tests it with a four times larger holdout

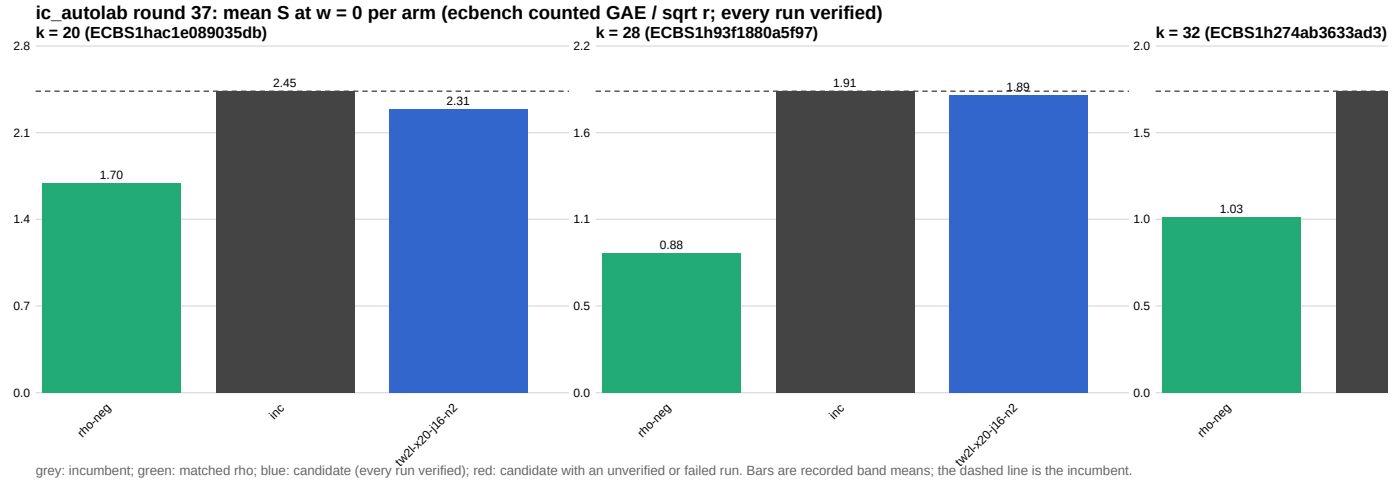
k = 28: session `ECBS1h93f1880a5f97` (run exit 0, verify exit 0, receipt sha256 `ace77c39993091d89bae957d69d6d6d0cb1ab4f0923e836432ed7dcea3df52ab`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	16	16	0.884 / 0.966 / 1.703	matched reference
inc	wwalkGg-x09-sr	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2,seed=run	16	16	1.912 / 2.093 / 3.726	incumbent
tw2l-x20-j16-n2	tw2l-x20-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.885 / 2.068 / 3.709	the SNC-panel arm (PCC-20261110-978c29: 0.936 / 0.909 / 0.922 of the incumbent at k = 20 / 28 / 32 over 480 multi-curve pairs); round 36's 64-pair holdout left these sizes null (0.998 / 0.958 / 0.942 with intervals touching 1), so this round re-tests it with a four times larger holdout

k = 32: session `ECBS1h274ab3633ad3` (run exit 0, verify exit 0, receipt sha256 `83480de0e162fda1f5d39ad71052d3d34103f7b1e5773806371b56be6ffbf689`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	16	16	1.031 / 1.133 / 2.046	matched reference
inc	wwalkG-x06-sr	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.6,g=1,grow=2,seed=run	16	16	1.773 / 2.000 / 4.041	incumbent
tw2l-x20-j16-n2	tw2l-x20-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.707 / 1.876 / 3.395	the SNC-panel arm (PCC-20261110-978c29: 0.936 / 0.909 / 0.922 of the incumbent at k = 20 / 28 / 32 over 480 multi-curve pairs); round 36's 64-pair holdout left these sizes null (0.998 / 0.958 / 0.942 with intervals touching 1), so this round re-tests it with a four times

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							larger holdout



grey: incumbent; green: matched rho; blue: candidate (every run verified); red: candidate with an unverified or failed run. Bars are recorded band means; the dashed line is the incumbent.

train screen

Selection

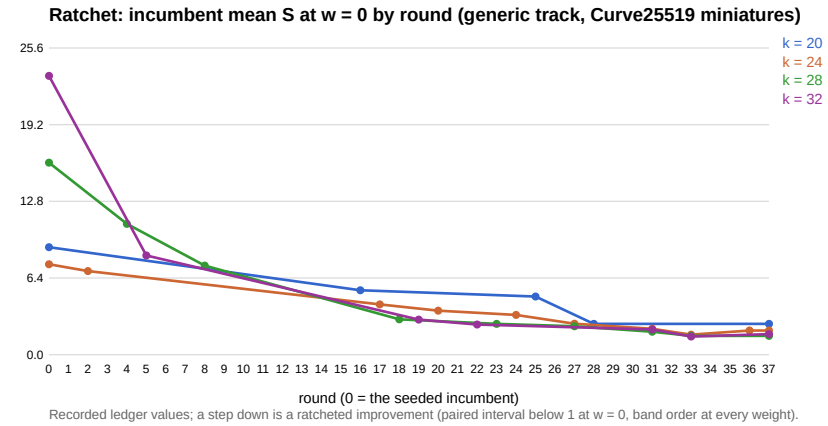
- k = 20: challenger `tw2l-x20-j16-n2` (Pareto alternative at w = 1: `tw2l-x20-j16-n2`)
- k = 28: challenger `tw2l-x20-j16-n2` (Pareto alternative at w = 1: `tw2l-x20-j16-n2`)
- k = 32: challenger `tw2l-x20-j16-n2` (Pareto alternative at w = 1: `tw2l-x20-j16-n2`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
20	`ECBS1h26723704be6c`	0	0	`19fa4b43810d94cd6f676f98bf090d076d5a74628bdd09c62996a101a40bfbdf`	{“verified”:1600}
28	`ECBS1hc56a473e04fb`	0	0	`ac9c4f7bc68afee944b2051ccb70e6b7ff2d362bcbe44f2541e0ac235231941f`	{“verified”:1600}
32	`ECBS1h879b9a8f4677`	0	0	`6eef034fb7e60d80946bc3d7c95acaa129ab59adaa71c798c2ed2b910e45ef91`	{“verified”:1600}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum2g-b15-n2` (`prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=15,free=1`)	28	2.575 / 2.691 / 3.741	1.8x / 1.7x / 1.5x
24	`tw2l-x20-j32-n2` (`prime-projected-abscissa:size=2,scan_cap=32 + wwalk:x=2.0,jumps=32,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run`)	36	2.016 / 2.189 / 3.743	1.8x / 1.8x / 1.8x
28	`wwalkGg-x09-sr-n2` (`prime-projected-abscissa:size=2,scan_cap=32 + wwalk:x=0.9,g=1,grow=2,seed=run`)	33	1.883 / 1.926 / 2.309	1.8x / 1.7x / 1.1x
32	`tw2l-x20-j16-n2` (`prime-projected-abscissa:size=2,scan_cap=32 + wwalk:x=2.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run`)	37	1.715 / 1.884 / 3.411	1.8x / 1.8x / 1.8x



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `1163f6b35d27ec8853455f6d88dfb9fb1d824745068df4dfd065bf6ada23b859`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.